

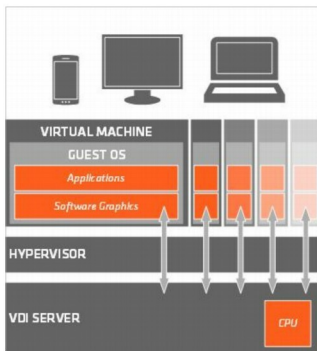


Figure 1: Technological overview of a virtual machine

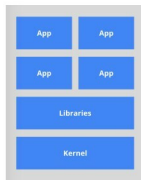


Figure 2: Heavyweight and non-portable

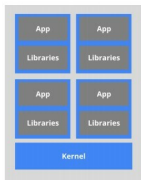


Figure 3: Lightweight and portable

decades later, we have containers that are a speedier, more efficient construct, which use operating system-level virtualisation instead. Containers offer a number of advantages over virtual machines, an important one being that they are lightweight and easier to deploy at scale in cloud-based infrastructure.

Docker

Docker is software that offers a means to develop and maintain containers. It primarily relies on the LXC interface and uses the concepts of groups and namespaces in Linux; however, the team has developed a programming interface called *libcontainer*, which is being used to develop current iterations.

The most important reason behind the high adoption rate of Docker is that it is lightweight. Docker images are much lighter to deploy than hypervisor-based systems like virtual machines.

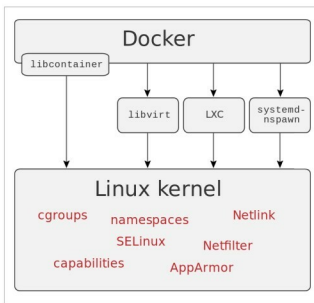


Figure 4: The Docker container

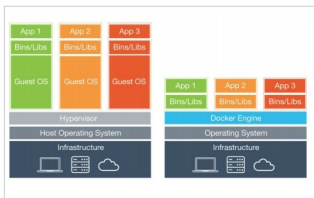


Figure 5: Comparing VMs with containers

This also increases the density of containers (compared to VMs) that can be deployed on a host. And this means that the size of a container is a few megabytes versus the gigabytes of a VM, and the boot time is in seconds contrasted with the minutes it takes for even the faster VMs to launch.

Docker Hub has gone a step further by providing a complete ecosystem for sharing and distributing images as well as maintaining existing deployments.

Enterprise usage

The application life cycle is a labour-intensive circle of development, integration and testing. Having a containerised environment greatly simplifies the dependency management, scalability and deployment of applications, since the deployment environment can be made completely identical across development and production servers. This is why Docker's simplification of complex development environments makes it a leading product across the industry.